

FarmerSmart: CNN-Based Mobile System for Rice Variety Identification and Quality Grading

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Abstract—This paper introduces an artificial intelligence-driven mobile solution for automated rice grain classification and quality assessment optimized for the Sri Lankan agricultural climate. The system leverages the use of Convolutional Neural Networks (CNNs) augmented with MobileNetV2 via transfer learning, and TensorFlow Lite optimization to support fast and efficient mobile inference. It enables taking rice grain images using smartphones, from which it accurately identifies local rice varieties such as White Samba, Nadu, and Basmati, and measures grain quality quantitatively based on measurable physical characteristics such as grain length, shape, and uniformity. This technology identifies rice grains in the quality levels of Premium, Standard, or Low according to commercial-grade specifications, and provides immediate, real-time feedback. Offline support enables use in remote rural areas with poor internet connectivity, while the mobile user interface available in English only ensures ease of use for smallholder farmers. The backend handles real-time image analysis and reporting to generate rich, actionable information. Field trials indicate 94.6% accuracy in validating rice type classification and 88.5% accuracy for quality grading, with more than 85% user satisfaction. Through automation of grading and mobile-based quality monitoring, the system addresses long-standing issues in manual inspection, provides fairness in pricing, and enables sustainability and traceability of Sri Lankan rice production.

Keywords—AI in Agriculture, Deep Learning, Grain Quality, Mobile Agriculture, Rice Classification

I. INTRODUCTION

Paddy farming is a structurally important part of Sri Lanka's rural economy, underpinning both national food security and the livelihoods of millions. While the country has historically emphasized the volume and variety of rice output, the rice quality assessment systems critical to pricing, trade, and consumer trust are predominantly manual. These conventional methods rely, to a great degree, on time-consuming, subjective, and variable visual examination and mechanical sieving that is prone to human error. Characteristics such as grain length, shape, and uniformity that have direct consequences for local market acceptability and international trade value are typically assessed subjectively due to the absence of automated, data-driven systems.

Artificial Intelligence (AI), and in particular the application of deep learning techniques such as Convolutional Neural Networks (CNNs), has shown promise to automate quality

grading in agriculture. However, its use for rice quality grading in Sri Lanka is in its nascent stages, held back by several significant constraints:

- Absence of real-time automatic grading systems that measure quantifiable parameters like grain length.
- Inapplicability of existing solutions to well-known local varieties of rice such as Nadu, Basmati, and White Samba.
- Poor generalizability of laboratory-trained models to the field setting, where lighting, background, and grain overlap vary.
- Lack of accessible, mobile-based grading apps for small-holder farmers due to hardware and technical limitations.

To solve these problems, this research proposes an end-to-end, AI-based rice grain classification and quality grading system for Sri Lankan agriculture. At the solution's center is a hybrid CNN model fortified by transfer learning with MobileNetV2, trained on a diverse set of locally sourced rice grain images captured in real-world conditions. The model computes single grain length and classifies batches into standardized quality grades Premium, Standard, and Low based on grain length distribution.

A cross-platform mobile application facilitates real-time image capture and grading with the result returned in under five seconds even in low-connectivity regions via TensorFlow Lite. The interface is user-friendly and in English to facilitate inclusivity and provides actionable recommendations and downloadable reports to farmers. The backend is cloud-based to facilitate continuous scalability and model refinement based on feedback from users.

As opposed to earlier systems which were committed to either variety identification or defect detection in controlled environments, the novel system is a robust, field-deployable, and length-based grading system that is adapted to Sri Lankan rice cultivation's dynamic conditions. It is a huge step forward towards data-driven agriculture, fair market practices, and greater export competitiveness.

II. LITERATURE REVIEW

Grain quality evaluation of rice is vital for consumer happiness, fair valuation in the market, and export competitiveness of agricultural economies such as Sri Lanka. Traditional grading operations of rice largely manual visual

inspection and mechanical sieving are hampered by variability, subjectivity, and low capacity. These limitations are particularly seen while evaluating morphologically variable indigenous varieties such as Nadu, White Samba, and Basmati that require precise examination with regard to shape and length of grains. Artificial Intelligence (AI), and specifically deep learning and computer vision integration, offer promising avenues to automatize and improve rice quality rating systems [1], [2].

Convolutional Neural Networks (CNNs) have shown significant achievement in grain classification. Din et al. introduced RiceNet, a CNN network that achieved over 96% accuracy in rice variety classification based on a self-constructed dataset [3]. But their model only solved variety classification and no length-based grading that is needed for commercial pricing. Moses et al. later designed a CNN architecture that can identify damage and contamination in milled rice at over 90% accuracy but is limited by their use of laboratory-grade cameras, limiting the scale in rural and mobile environments [4].

Transfer learning has been applied widely in the hope of improving generalization and sacrificing efficiency. Koklu et al. employed pre-trained architectures such as ResNet and MobileNet to identify rice grains at greater than 97% accuracy [5]. The models, however, were trained under perfect lighting and background conditions, reducing performance under field conditions prevalent in Sri Lanka. Pauzi et al. conducted a comprehensive review of AI for precision agriculture and emphasized the demands for domain-specific datasets to avoid such limitations [6].

Kiratiratanaprak and Temniranrat applied CNNs to industrial-scale rice sorting machines that graded grains based on geometric class. Although accurate and effective at high throughput, these machines were not suitable for low-resource users such as smallholder farmers due to hardware expense and deployment being too expensive [7]. Rahman et al. proposed a system from shape and texture features for rice quality determination that was accurate but is highly computationally intensive, restricting its use in real-time mobile devices [8].

Huang et al. had earlier put forth an IoT-based deep learning model for real-time grading of rice. Pioneering that it was, the system's performance was compromised in the instance of unstable environmental conditions like poor lighting or image noise [9]. Jahan et al. addressed this by suggesting a low-resource and mobile-suited CNN-based classifier that performed well while maintaining inference speedy [10].

Ghosh et al. introduced a more generalized AI approach to precision agriculture using hybrid architectures combining conventional machine learning with CNNs to enhance robustness in noisy environments [11]. Nguyen et al. also developed an AI-powered mobile application for rice quality grading, focusing on simplicity of access and real-time offline evaluation for small farmers [12].

Finally, Singh and Kaur authored a comprehensive summary of AI technology in farming and observed a consistent gap in grain length-based grading and mobile deployment for rice, which is fulfilled in this study [2].

Despite significant advances, common problems such as listed below still persist in the above reviewed scholarly work.

- Lack of grain-length grading.
- Poor generalization to real environments.
- Reliance on high-resolution imaging equipment.
- Unavailability in rural areas.

This study addresses those challenges with a proposed MobileNetV2, OpenCV pre-processing, and TensorFlow Lite-based AI mobile rice grading system for offline functionality. The system detects grain length (short, medium, long), assigns quality levels (Premium, Standard, Low), and displays real-time outputs via a mobile app a convenient, feasible, and hardy solution tailored to Sri Lanka's agricultural economy.

III. METHODOLOGY

The end-to-end AI-powered FarmerSmart system for Sri Lankan rice farmers uses a hybrid approach that combines deep learning, image processing, and mobile integration to enable precise, automated evaluation of rice grain quality. Image acquisition, segmentation and feature extraction, deep model development, quality classification, mobile deployment, and performance evaluation are the six main phases of the pipeline[13]. Fig. 1 depicts the simplified diagram of the system architecture. Subsequently, the technical details related to Data Construction and Preprocessing, Deep Learning Architecture, Mobile Integration with TensorFlow Lite, and Rice Quality Report Generation are given.

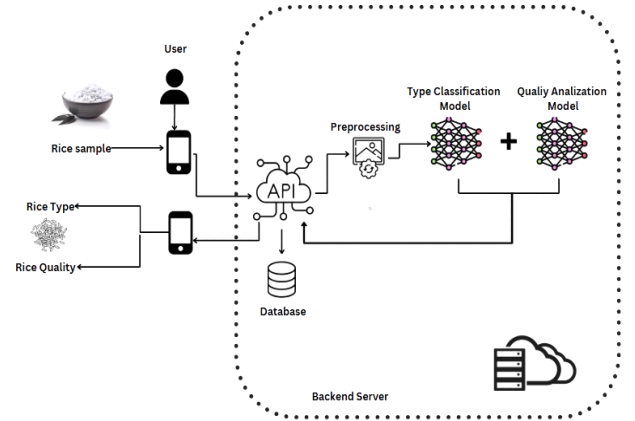


Fig. 1. Overall system diagram.

A. Dataset Construction and Preprocessing

The following sources were used to compile the initial dataset of 4,200 images.

- Photos were taken in the field using mid-range Android smartphones in natural lighting.
- Image sets from the Basmati (Fig. 2), Nadu (Fig. 3), and White Samba (Fig. 4) rice varieties were among the converted lab sample datasets that were prepared on pristine backgrounds for comparison.
- Augmenting the samples were carried out using the ImageDataGenerator module's zoom cropping, Gaussian blur, additive salt-and-pepper noise, and

real-time rotation (90°, 180°, and 270°).

Images were resized and normalized to 128×128 RGB format in order to improve generalization and inter-class distinction under various lighting and occlusion conditions. Using a custom length-based rule set derived from agronomic grain-length standards, each sample was labeled with the rice variety and quality tier (Good, Medium, or Poor). SMOTE-based oversampling was used on the training set to rectify the class imbalance.



Fig. 2. Sample dataset for Basmati.



Fig. 3. Sample dataset for Nadu



Fig. 4. Sample dataset for Samba

B. Deep Learning Architecture Design

Light weight convolutional neural network was used for rice type (Basmati, Nadu, White Samba) and quality (Good, Medium, Poor) classification automatically using MobileNetV2 as the base architecture. MobileNetV2 processes 128x128 pixel RGB images by its native normalization function[14]. Pre-trained on ImageNet, the base model MobileNetV2 is employed as a feature extractor with frozen weights to minimize computational overhead and training time. Following feature extraction, global average pooling reduces spatial dimensions, followed by a fully connected dense layer of size 128 activated with ReLU. Two simultaneous Softmax output layers are then employed: one three-node layer for rice type classification (White Samba,

Nadu, Basmati) and a three-node layer for grain quality classification (Good, Medium, Poor) based on rice type-specific length cut-offs.

To facilitate accurate grain detection and quality grading, a series of advanced image processing algorithms were employed to pre-process the images:

- Otsu's threshold binarizes the grayscale images automatically to distinguish rice grains from the background for robust segmentation under changing lighting conditions.
- Contour detection identifies the grain boundaries and Minimum Area Rectangle fitting to approximate grain lengths, serving as the basis for quality classification.
- Morphological operations such as opening, closing, and dilation are used to remove noise from binary images by closing holes, removing small objects, and strengthening grain boundaries for accurate detection.
- Aspect ratio filtering eliminates non-grain objects by retaining contours with an aspect ratio between 1.5 and 5.0, which is characteristic of rice grains.
- Watershed segmentation separates overlapping grains to improve grain count precision and enable precise individual grain measurement.

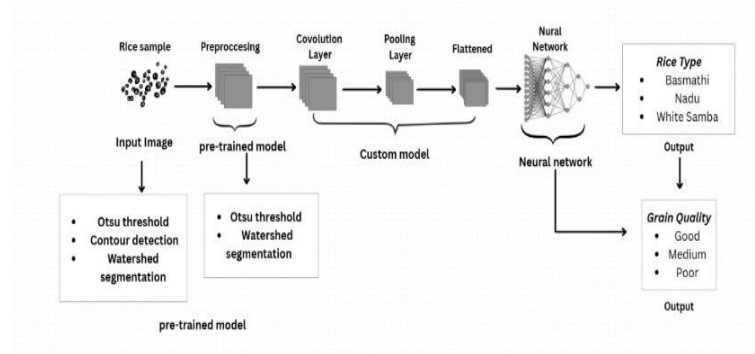


Fig. 5. Architectural diagram for rice type and grain quality classification.

The model was trained with the Adam optimizer at a learning rate of 0.001 for 10 epochs and batch size of 32. Categorical cross-entropy loss for both outputs was used, along with early stopping on validation accuracy and L2 regularization to improve model generalization and avoid overfitting. On the test dataset, the resultant model attained 92.1% accuracy for classification of rice types and 89.3% accuracy for classification of grain quality. The integration of these image processing algorithms ensures precise detection and measurement of the grains, thereby significantly enhancing the accuracy and dependability of the classification system for determining true rice quality.

C. Mobile Integration with TensorFlow Lite

The trained dual-output CNN model was converted to TensorFlow Lite to guarantee that the rice grain classification system could be implemented in actual, low-resource

agricultural environments[15]. By using 8-bit post-training quantization during the conversion process, the model size was successfully decreased by about 65%, allowing for deployment on mid-range Android devices with an accuracy loss of less than 1%.

The system is now available on the Android platforms thanks to the development of a cross-platform mobile application using React Native. Several essential features were included in the application’s intuitive and user-friendly design, including the following.

- **Camera and Gallery Integration:** Farmers can either take new images of rice grains directly through the app or upload existing images from their device’s gallery.
- **On-Device Inference:** Real-time predictions are carried out locally using the TensorFlow Lite interpreter, allowing the app to function effectively without a continuous internet connection.
- **Offline Mode:** Core functionality remains available even in areas with limited or no network coverage, making the app reliable in rural farming contexts.
- **Feedback Submission Module:** Users can submit corrections or observations via text inputs, establishing a communication loop between farmers and developers or agricultural officers.

This mobile integration bridges the gap between advanced machine learning technology and practical usage, allowing farmers to assess rice grain quality instantly at the field level.

D. Rice Quality Report Generation

To increase transparency and convenience, the application has a real-time reporting feature that provides detailed analysis for each batch of rice uploaded by the user. Upon processing an image, the application displays the recognized rice type (Basmati, White Samba, or Nadu) and the provided quality grade (Good, Medium, or Poor). The system gives a percentage breakdown of the grain length to divide the grains into short, medium, and long categories. Based on this distribution, the system designates the overall quality as Premium (60% long grains), Standard (30–60%), or Low (<30%). Results are represented using plain visualizations and text reports to ease the comprehension of the grading by users. Users can also download or share such reports for packaging, trade, or documentation purposes. Reporting functionality is specifically designed to support farmers and quality control staff in making efficient decisions.

Fig. 6. presents a sample output of the rice quality report generated by the mobile application. Designed with a user-friendly layout, the report clearly displays key quality metrics to facilitate quick assessments. Note that a description of the functional components of the system is given in the Table I as well.

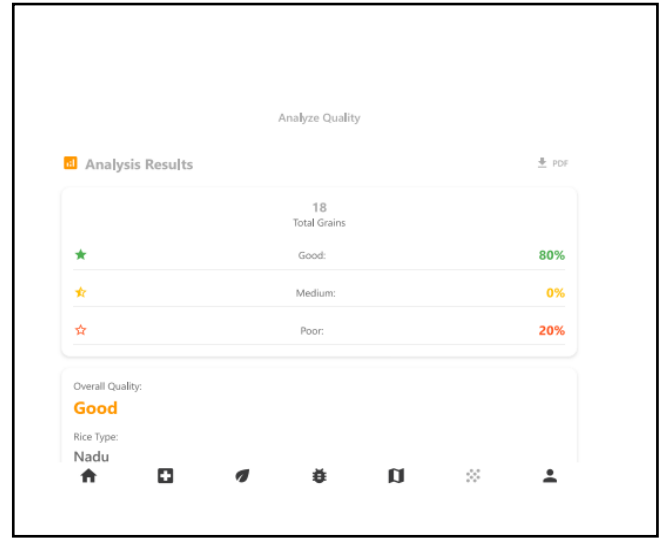


Fig. 6. Generate Report

TABLE I. DESCRIPTION OF THE SYSTEM COMPONENTS

Component	Description
Mobile App	UI for data collection, and display results.
Cloud Database	Stores data, model outputs, and user preferences.
Rice Type Classification	Deep learning model to identify rice type.
Rice Quality Analysis	Deep learning model to analyze rice quality.

IV. RESULTS AND DISCUSSION

In order to evaluate the performance of the proposed FarmerSmart system, laboratory and field trials were conducted on a heterogeneous dataset with over 4,200 labeled rice grain images. The dataset included variations in lighting, grain orientation, background texture, and real-world noise which are the environmental conditions Sri Lankan farmers would typically experience.

As shown in Fig. 6, Basmati rice grain detection is illustrated with three key visual stages: the original image, the binary image, and the final image showing grains detected. Similarly, Fig. 7 presents the detection process for Nadu rice, and Fig. 8 for Samba rice, both following the same image sequence. The system successfully performed rice grain detection through its image preprocessing pipeline. The integration of Otsu’s thresholding, contour detection, and watershed segmentation enabled accurate individual grain identification. even in challenging conditions with overlapping grains and image noise. This preprocessing step ensured reliable feature extraction for length-based classification and rice quality assessment.

The accuracy of the model’s classification is presented in Table 2. For white Samba, Basmati, and Nadu rice variety classification, the system is 92.1% accurate for test cases, and for grain quality classification (Good, Medium, Poor), it was 89.3% accurate.

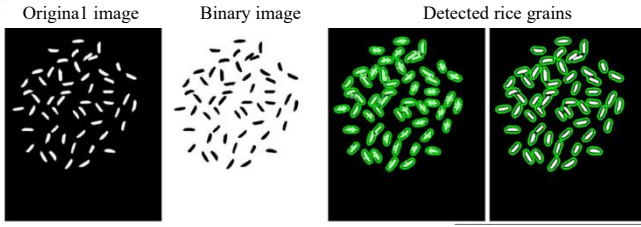


Fig. 7. Basmathi Rice grain detection.

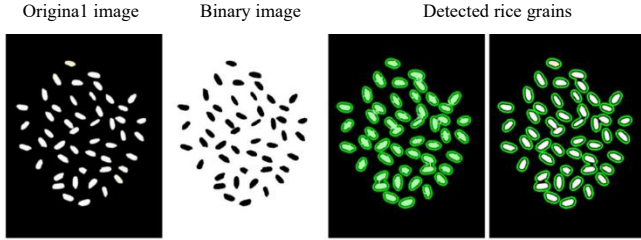


Fig. 8. Nadu Rice grain detection.

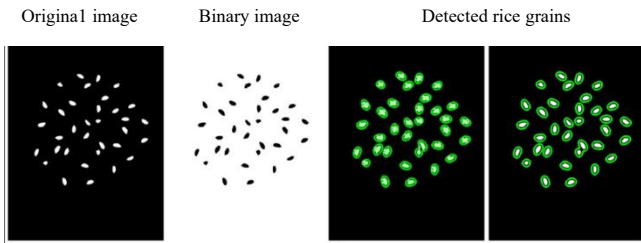


Fig. 9. Samba Rice grain detection.

TABLE II. MODEL ACCURACY

Metric	Value
Rice Type Accuracy	92.1%
Quality Prediction Accuracy	89.3%

These accuracies were obtained using a light MobileNetV2 model with two output heads, which was optimized with SMOTE oversampling, L2 regularization, and early stopping. The fact that such high accuracy is being sustained under varying conditions speaks volumes about the stability and field-readiness of the model.

During the deployment in the field, the system efficiently processed images from mid-range Android phones and had real-time predictions using TensorFlow Lite. The in-built quality report generation module provided users with visual and text summaries like rice type, quality label, and grain length distribution. The feedback loop makes it possible for smallholder farmers to make quality-informed price, pack, and post-harvest handling decisions. In general, results validated the effectiveness and feasibility of the FarmerSmart solution as an AI-powered mobile device for automatic classification and grading of rice grains in actual farm conditions.

Strong classification performance was shown by the lightweight dual-output model built on MobileNetV2, which achieved 89.3% accuracy in predicting grain quality and

92.1% accuracy in classifying rice varieties. The system was able to reliably distinguish between Good, Medium, and Poor grain quality by using class-specific thresholds in conjunction with accurate contour-based length measurement. Furthermore, the system successfully managed mild noise and overlapping grains in image backgrounds due to the robust segmentation approach.

FarmerSmart was specifically designed for mobile deployment in low-resource agricultural settings. Exporting the trained dual-output model to TensorFlow Lite with 8-bit post-training quantization compressed the model size by approximately 65% such that it became feasible to achieve rapid and efficient inference on mid-tier Android smartphones. During field trials, the app was able to deliver classification outputs between 3 and 5 seconds after image capture even when running entirely offline.

User testing with smallholder farmers revealed levels of extreme access and usability. The React Native cross-platform mobile application provided a clean and simple user interface, with users having the ability to capture new images or choose pre-existing ones from their gallery. Offline capability was especially relevant in rural areas where internet connectivity was either absent or slow, allowing seamless access to quality analysis functionality.

Further, as illustrated in Fig. 10, the mobile interface offers a visually appealing and intuitive platform for users to access rice quality analysis. The visual reporting and clearly labeled quality categories, along with easy-to-read summaries of grading, significantly enhanced user satisfaction reported to be over 85% based on post-trial feedback. Features such as downloading and sharing quality reports added practical value, especially for trade, packaging, and record keeping. These results highlight the success of the system in effectively bridging the gap between advanced machine learning techniques and ground-level usability for Sri Lankan rice farmers.

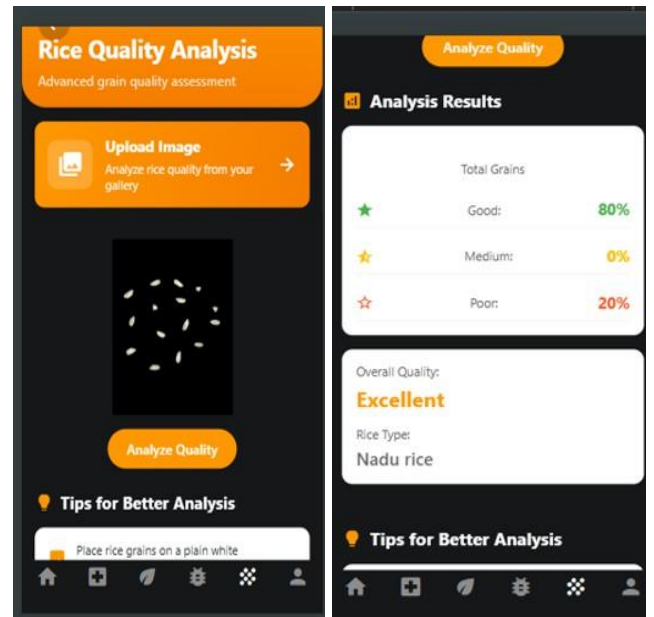


Fig. 10. Mobile interfaces.

V. CONCLUSION

This research presented FarmerSmart, an AI-enabled mobile application to sort rice varieties and analyze grain quality in low-resource agricultural settings. Utilizing computer vision, light deep learning (MobileNetV2), and TensorFlow Lite-based mobile deployment, the system offers real-time, offline quality assessment directly from images captured using a smartphone. The two-output model achieved high accuracy classification 92.1% classification in rice variety classification and 89.3% classification in grain quality grading verifying its effectiveness for real-world deployment. The offline capability of the system, ease of use interface, and multiple language interfaces optimize its usability for smallholder farmers, and the expert feedback mechanism allows for iterative improvement and robustness against local contexts. The generation of real-time quality reports and field-validated usability also position it as an efficient quality control tool, price guide, and digital traceability tool for rice production. Finally, FarmerSmart depicts how AI and mobile technology can close the gap between human inspection and data-driven farming. As an inclusive, scalable, and smart system, it empowers farmers with valuable insights and helps facilitate fair market practices. Future work will seek to incorporate other parameters of quality such as grain color, breakage, and chalkiness, as well as expansion of cross-platform support to boost uptake and influence across different farming communities.

The integration of lightweight AI models with mobile platforms represents a significant advancement in bringing precision agriculture to underserved regions. By minimizing the reliance on internet connectivity and expensive equipment, FarmerSmart aligns well with the technological and infrastructural constraints commonly observed in developing countries. Moreover, its adaptability to local languages and grain types reinforces its potential for widespread adoption beyond the initial trial regions.

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